Module 6 - One-Page Retrospective

Group 19 | Real-Time Network Intrusion Detection | FUSE AI-201

Group 19 progressed from problem identification and model development to a publicly accessible research prototype. The project combines a five-class Random Forest classifier, an NSL-KDD feature pipeline, a Streamlit CSV interface, an analytics dashboard, source control, automated software tests and a proposed operational maintenance plan.

Achievements

We developed a reproducible training/export workflow using public NSL-KDD KDDTrain+ and KDDTest+ records, consolidated labels into Normal, DoS, Probe, U2R and R2L, and integrated SMOTENC within the training pipeline. We selected a Random Forest pipeline and published the source repository and model artifacts. The public Streamlit application accepts a 41-feature CSV, generates class predictions and downloadable results, and includes benchmark analytics. The final source package passed 29 automated tests in the finalization environment.

Challenges

The largest technical and research challenge was the gap between validation and independent KDDTest+ performance. The validation Macro F1 was 98.26%, while independent-test Macro F1 was 51.51%. The team also addressed deployment and dashboard issues, including a hosted Plotly false-positive-rate chart error; a correction and table fallback were committed to GitHub as 8941226. The final Cloud rebuild after that hotfix still needs a smoke-test confirmation.

Lessons learned

Keeping preprocessing and SMOTENC inside the model pipeline helps maintain a consistent training process and avoids oversampling validation/test data. An untouched independent test set is essential: high validation scores do not guarantee generalization. Class-wise recall and confusion patterns are necessary because overall accuracy can conceal missed attack categories. A displayed maximum class probability should not be called calibrated confidence without a separate calibration study.

Limitations

Independent-test recall is 8.96% for U2R and 7.20% for R2L. NSL-KDD is a historical benchmark and does not establish performance on modern enterprise traffic or zero-day attacks. The hosted app classifies uploaded connection records; it does not capture live packets, block traffic or provide continuous production monitoring. Monitoring, drift response, shadow deployment, fallback IDS and retraining processes are proposed plans, not active services.

Planned improvements

Future work should investigate U2R/R2L false negatives, test class-sensitive objectives and calibration, use newer authorized datasets with verified labels, maintain a separate final test set, and conduct a controlled pilot with analyst oversight and privacy/security safeguards. Before submission, the group will confirm the final Streamlit Cloud hotfix deployment, complete individual contribution records, and record the required video.

GitHub: https://github.com/adugnaNecho/Group19-network-intrusion-detection
Streamlit: https://group19-network-intrusion-detection.streamlit.app/